

Table of Contents

1. Introduction	3
2. Problem Statement	3
3. Project Objectives	4
4. System Architecture	5
5. System Components.....	6
6. Technology Stack.....	9
7. Known Limitations.....	10
8. Future Improvements.....	10
9. Conclusion	11

Table of Figures

Figure 1: UI	11
Figure 2: Inputs	12
Figure 3: Output	13
Figure 4: CSV File	14

Resume Screening System

1. Introduction

Recruitment processes often involve reviewing a large number of resumes manually, which can be time-consuming and inconsistent. This project focuses on developing an AI-based Resume Screening System that helps automate the initial candidate evaluation process.

The system analyzes multiple candidate resumes and compares them with a given job description. It extracts relevant skills, calculates a matching score, identifies missing skills, generates candidate summaries, and ranks candidates based on their suitability for the role.

The application is built using Python and Streamlit, with machine learning techniques and AI-powered analysis to provide a faster and more structured screening process.

2. Problem Statement

Hiring teams receive many applications for a single job position, making it difficult to manually analyze every resume in detail.

The main challenges are:

- Time-consuming manual resume screening
- Difficulty identifying relevant skills quickly
- Inconsistent candidate evaluation
- Lack of automated ranking between candidates

This project solves these problems by creating an automated system that evaluates resumes based on job requirements and provides meaningful insights about each candidate.

3. Project Objectives

The main objectives of this project are:

- Develop an AI-based resume analysis application
- Extract candidate information and skills from PDF resumes
- Compare candidate skills with job requirements
- Calculate a candidate-job matching score
- Identify missing skills for each candidate
- Rank candidates based on suitability
- Generate AI-based candidate summaries
- Provide downloadable screening results

4. System Architecture

The system follows the following workflow:

Multiple PDF Resumes



PDF Text Extraction

(pdfplumber)



Skill Extraction

(Groq API - Llama 3.3 70B)



Resume and Job Description Comparison



Scoring Engine

(TF-IDF + Skill Matching)



Candidate Ranking



AI Summary Generation



Streamlit Dashboard



CSV Export

5. System Components

5.1 PDF Resume Processing

The system accepts multiple PDF resumes as input.

The pdfplumber library is used to extract readable text from uploaded PDF files.

Process:

1. User uploads resume PDFs
2. Text content is extracted from each page
3. Extracted text is passed for skill analysis

This allows the system to work directly with real-world resume documents.

5.2 Skill Extraction

After extracting resume text, the system uses the Groq API with Llama 3.3 70B model to identify important candidate skills.

The AI model analyzes the resume content and extracts technical skills related to the job requirements.

Examples:

Python

Machine Learning

Streamlit

Pandas

Scikit-learn

SQL

This approach allows the system to understand skills even when they appear in different formats inside resumes.

5.3 Resume Matching and Scoring

Candidate ranking is calculated using two methods:

Skill Matching Score (60%)

The system compares candidate skills with required job skills.

Formula:

Matched Skills / Total Required Skills × 100

Example:

Required skills:

Python

Machine Learning

SQL

AWS

Candidate skills:

Python

Machine Learning

SQL

Skill score:

$3/4 \times 100 = 75\%$

TF-IDF Similarity Score (40%)

The system also compares the overall text similarity between:

- Resume content
- Job description

using TF-IDF vectorization and cosine similarity from Scikit-learn.

This helps identify candidates whose experience and background are similar to the job requirements.

Final Score

The final matching score is calculated as:

Final Score =

$(0.6 \times \text{Skill Score})$

+

$(0.4 \times \text{TF-IDF Similarity Score})$

Candidate categories:

Score	Recommendation
70% and above	Recommended
40% - 69%	Consider
Below 40%	Not Suitable

5.4 AI Candidate Summary

For each candidate, the system generates a short professional summary using the Groq AI model.

The summary includes:

- Candidate strengths
- Relevant experience
- Important skills
- Possible skill gaps

Example:

The candidate has strong Python and machine learning experience with practical projects in Streamlit and data analysis.

Additional knowledge of cloud technologies would improve suitability.

5.5 Candidate Ranking

After processing all resumes, candidates are automatically ranked from highest matching score to lowest.

The ranking table provides:

- Candidate name
- Matching score
- Recommendation label
- Matched skills
- Missing skills

This helps recruiters quickly identify suitable candidates.

6. Technology Stack

Python

Used as the main programming language for application development and data processing.

Streamlit

Used to create the interactive web interface for uploading resumes and displaying results.

pdfplumber

Used for extracting text from PDF resumes.

Groq API (Llama 3.3 70B)

Used for AI-based skill extraction and candidate summary generation.

Scikit-learn

Used for:

- TF-IDF vectorization

- Cosine similarity calculation

Pandas

Used for:

- Managing candidate data
- Creating ranking tables
- Exporting CSV results

python-dotenv

Used for securely managing API keys through environment variables.

7. Known Limitations

Although the system provides automated screening, some limitations exist:

- Resume formatting differences may affect text extraction
 - AI-generated summaries depend on model output quality
 - Matching accuracy depends on the provided job description
 - Complex professional experiences may require human review
 - The system focuses mainly on technical skill matching
-

8. Future Improvements

Possible improvements include:

- Adding more advanced NLP models
 - Supporting more resume formats
 - Adding experience and education weighting
 - Integrating a database for candidate history
 - Adding recruiter feedback learning system
-

9. Conclusion

The AI Resume Screening System provides an automated approach for evaluating candidates against job requirements.

By combining PDF processing, AI-based skill extraction, machine learning similarity scoring, and automated ranking, the system reduces manual screening effort and helps identify suitable candidates faster.

The project demonstrates how AI and machine learning techniques can be applied to improve recruitment workflows.

Demo Screenshots

The following screenshots demonstrate the working process of the application.

Screenshot 1: Job Description Input and Resume Upload UI

Deploy ⓘ

Resume Screening System

Upload resumes and a job description to rank and evaluate candidates automatically.

JOB DESCRIPTION

Paste the full job description here...

RESUME UPLOAD

Upload

200MB per file • PDF

Upload one or more PDF resumes. Each file name will be used as the candidate name.

Analyze Candidates

Figure 1: UI

Screenshot 2: Job Description Input and Resume Uploaded

Deploy

Resume Screening System

Upload resumes and a job description to rank and evaluate candidates automatically.

JOB DESCRIPTION

Job Title: AI/ML Developer Intern

We are looking for an AI/ML Developer Intern who has knowledge of Python, machine learning, and data science. The candidate will work on building machine learning applications, data analysis solutions, and AI-powered systems.

Responsibilities:

- Develop and implement machine learning models using Python
- Work with data preprocessing, analysis, and visualization
- Build AI applications using Streamlit and Python frameworks
- Extract and process data from different sources
- Integrate APIs and AI models into applications
- Evaluate model performance and improve accuracy
- Write clean and maintainable code

Required Skills:

- Python programming
- Machine Learning
- Scikit-learn
- Pandas
- NumPy
- Data Analysis
- Data Visualization
- Streamlit
- SQL
- Basic understanding of AI and ML algorithms
- Experience with APIs is a plus

Preferred Skills:

- Experience with Logistic Regression, SVM, Random Forest, XGBoost
- Knowledge of Natural Language Processing
- Experience building AI projects
- Familiarity with Git and GitHub

Education:
Bachelor's degree or currently pursuing a degree in Computer Science, Information Technology, Data Science, or related fields.

Candidate should be passionate about learning AI/ML and building practical solutions.

RESUME UPLOAD

CV_...pdf293.4KB

Latest_CV.pdf293.0KB

New_CV.pdf293.0KB

+

Upload one or more PDF resumes. Each file name will be used as the candidate name.

Analyze Candidates

Figure 2: Inputs

Screenshot 3: Ranked Candidate Results Table with AI Summary and Extracted Skills

Resume Screening System

Upload resumes and a job description to rank and evaluate candidates automatically.

JOB DESCRIPTION

* CAPABILITY: NLP, LOGICAL, REG. EXPERT, DATA, MATH, DATA SCIENCE, ANALYSIS

- Knowledge of Natural Language Processing
- Experience building AI projects
- Familiarity with Git and GitHub

Education:

Bachelor's degree or currently pursuing a degree in Computer Science, Information Technology, Data Science, or related fields.

Candidate should be passionate about learning AI/ML and building practical solutions.

RESUME UPLOAD

CV_...pdf 231.4KB Latest_CV.pdf 231.0KB New_CV.pdf 233.2KB +

Upload one or more PDF resumes. Each file name will be used as the candidate name.

Analyze Candidates

JOB REQUIREMENTS DETECTED

Python Machine Learning Data Science Scikit-learn Pandas NumPy Data Analysis Data Visualization Streamlit SQL APIs Logistic Regression SVM Random Forest XGBoost Natural Language Processing Git GitHub

PROCESSING CANDIDATES

Results — 3 Candidate(s) Ranked

No candidates scored above 70%. See full rankings below.

Rank	Candidate	Match Score (%)	Matched Skills	Missing Skills	Status
1	Latest Cv	43	11	7	Consider
2	New Cv	36.6	9	9	Not Suitable
3	Cv_..	35.8	9	9	Not Suitable

Candidate Details

▼ #1 — Latest Cv — 43.0% Match

MATCH SCORE

43.0%

MATCHED SKILLS (11)

Python Machine Learning Scikit-learn Pandas Data Analysis
Data Visualization Streamlit Logistic Regression SVM XGBoost
Natural Language Processing

MISSING SKILLS (7)

Data Science NumPy SQL APIs Random Forest Git GitHub

AI SUMMARY

Latest Cv is a candidate with a moderate match score of 43% whose skills align with requirements in Python, Machine Learning, and data analysis tools such as Scikit-learn and Pandas. Although they lack experience in key areas like Data Science, NumPy, SQL, APIs, and Random Forest, their existing skill set demonstrates potential for growth and development. Overall, Latest Cv may be a suitable candidate for roles that value foundational skills in Python and Machine Learning, with potential for further training and skill acquisition.

ALL EXTRACTED SKILLS FROM RESUME

Python Streamlit Groq API pdfplumber Pandas Plotly Logistic Regression Scikit-learn OpenMeteo API SVM XGBoost Matplotlib Seaborn JSON Natural Language Processing Machine Learning Data Analysis
Data Visualization

➤ #2 — New Cv — 36.6% Match

➤ #3 — Cv_.. — 35.8% Match

Download Results as CSV

Figure 3: Output

Screenshot 4: Candidate Detail Card

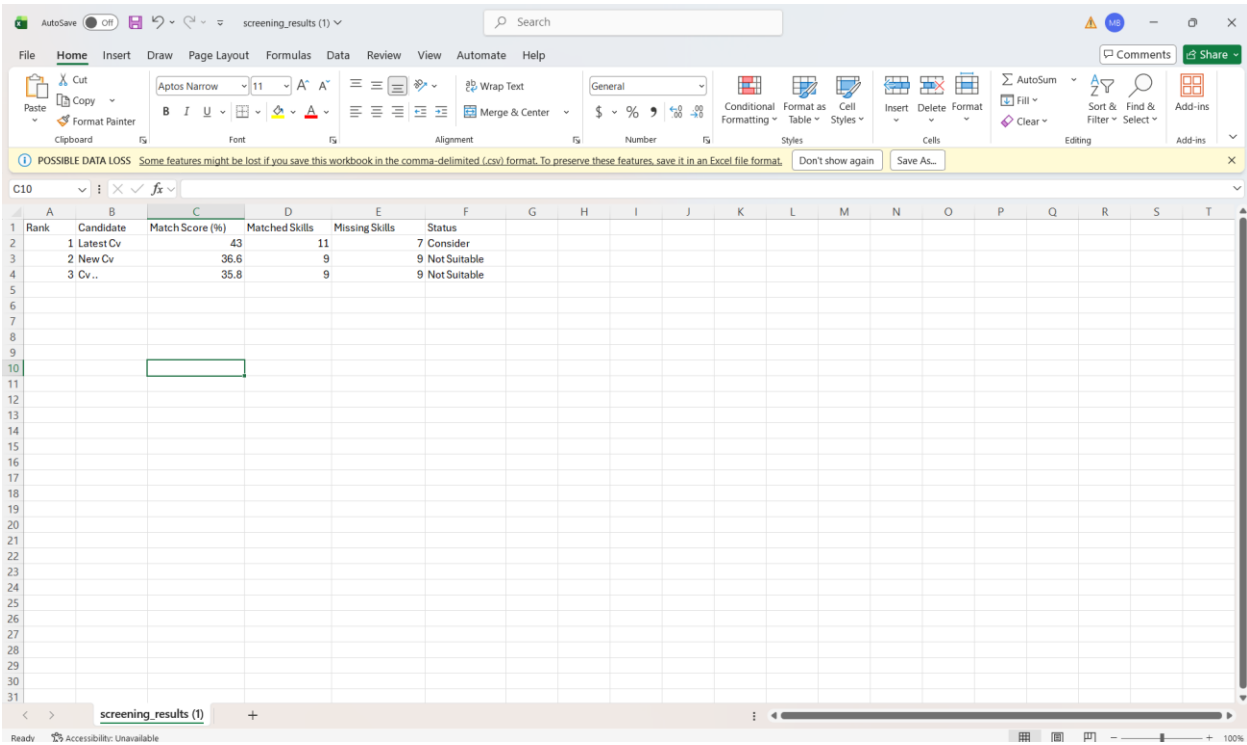


Figure 4: CSV File